

Course Materials & Exam Scope 1/2

Chapter 1 — Human aspects

expanded in lectures

book = good reference

Chapter 2 — Computer aspects

replaced by lectures (accessibility, assistive tech)

book = outdated → use Kiro + course materials

Course Materials & Exam Scope 2/2

Chapter 3 — Interaction

models & ergonomics → still valid

section 3.5 (interaction styles) → outdated

→ use lecture slides (updated + paradigms)

section 3.6 (wimp interfaces) → to be covered

and expanded → must be studied

Chapter 4

can be ignored

Interaction styles (Paradigms)

The dialog between the computer and the user

Traditional interaction styles

command line interface
menus
natural language
question/answer and query
dialogue
form-fills and spreadsheets
WIMP (window, icon,
menu, pointer)
point and click
gesture interfaces
three-dimensional interfaces

- Traditional interaction styles (from textbook)
- Still useful, but incomplete
- Today: more complex interaction landscape
- Goal: a clearer and modern structure

The Problem with the Old List

command line interface
menus
natural language
question/answer and query
dialogue
form-fills and spreadsheets
WIMP (window, icon,
menu, pointer)
point and click
gesture interfaces
three-dimensional interfaces

- Mixes different concepts:
 - paradigms
 - techniques
 - dialogue styles
- Some items overlap
- Hard to compare and organize

Let's try a new organization

We separate interaction into 3 levels:

Paradigms

→ overall interaction model

Interaction styles

→ how the dialogue works

Modalities

→ how users input and receive information

Why This Matters Today

Modern systems are **multimodal**

Different input/output channels
(touch, voice, gesture, gaze)

Modern systems use **multiple interaction styles**

Different ways of structuring interaction
(menus, direct manipulation, conversation)

Examples:

Smartphone = touch (modality) + menus (style)

AI systems (e.g. ChatGPT) = text (modality) +
conversation (style)

Interaction Paradigms

High-level models of interaction

They define:

- how users understand the system
- **the overall logic of interaction**

Different paradigms enable different styles and modalities

Interaction Paradigms We Will Cover

Classic paradigms

- Command Line Interfaces (CLI)
- Graphical User Interfaces (WIMP)

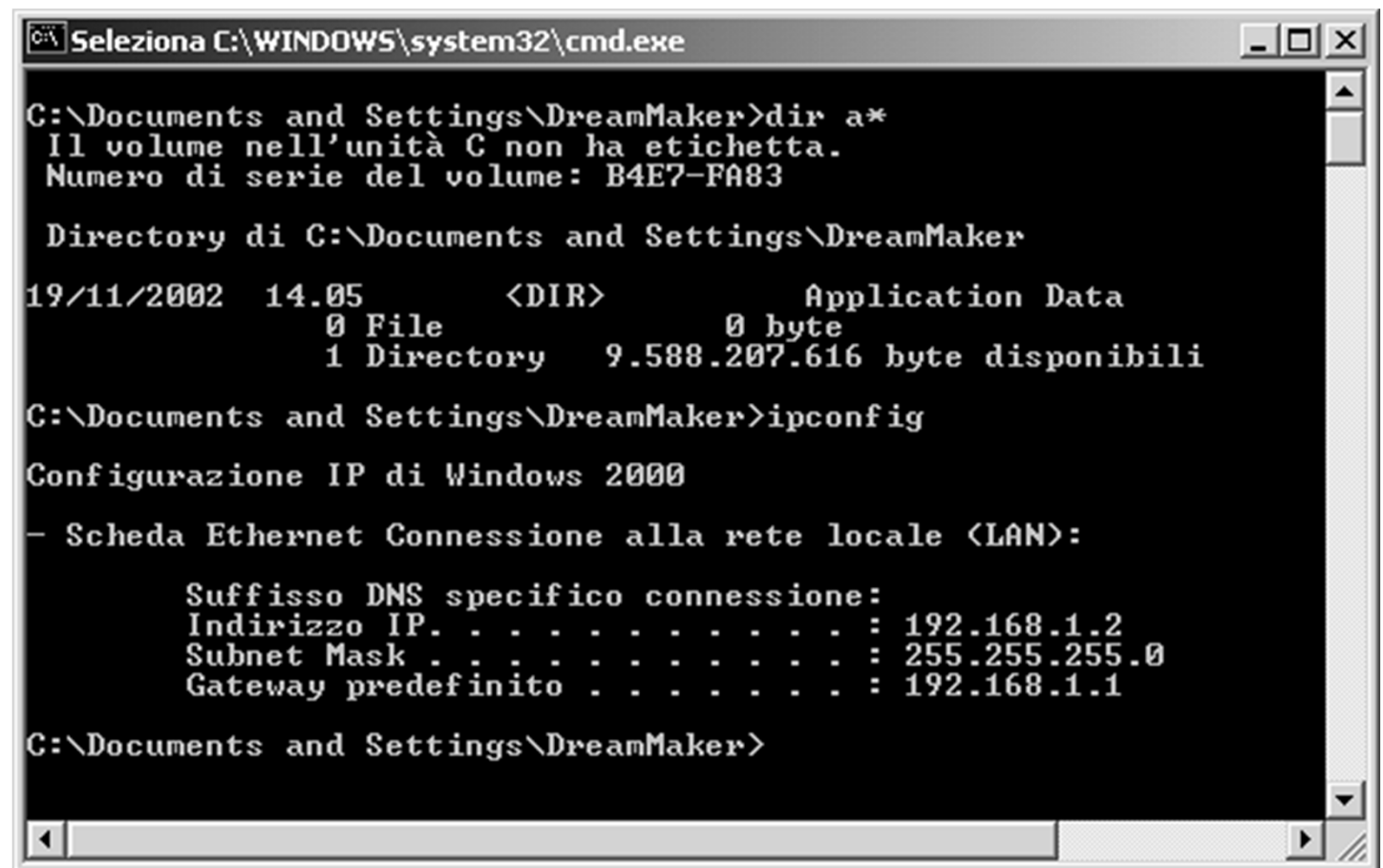
Modern paradigms

- Post-WIMP (touch, gesture)
- Conversational interfaces
- Immersive interfaces (AR/VR)
- Ubiquitous / ambient computing

Command Line Interface (CLI)

Way of **expressing instructions** to the computer **directly**

Function keys,
single characters, short abbreviations, whole words, or a combination (e.g. CTRL+C)



```
Seleziona C:\WINDOWS\system32\cmd.exe

C:\Documents and Settings\DreamMaker>dir a*
Il volume nell'unità C non ha etichetta.
Numero di serie del volume: B4E7-FA83

Directory di C:\Documents and Settings\DreamMaker
19/11/2002  14.05      <DIR>          Application Data
               0 File                0 byte
               1 Directory    9.588.207.616 byte disponibili

C:\Documents and Settings\DreamMaker>ipconfig

Configurazione IP di Windows 2000

- Scheda Ethernet Connessione alla rete locale (LAN):

    Suffisso DNS specifico connessione:
    Indirizzo IP. . . . . : 192.168.1.2
    Subnet Mask . . . . . : 255.255.255.0
    Gateway predefinito . . . . . : 192.168.1.1

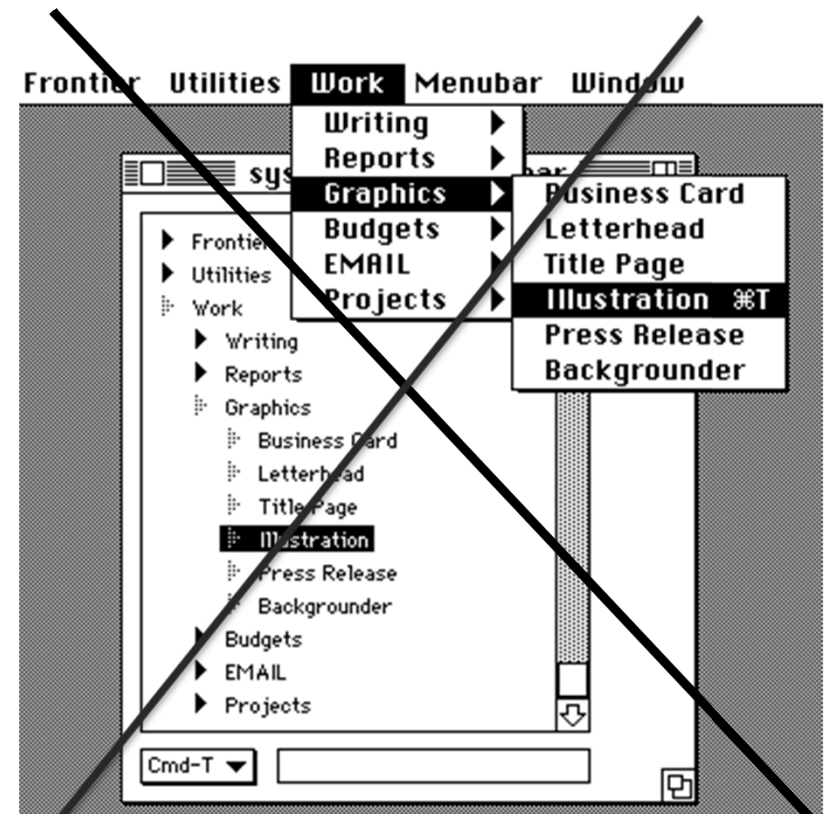
C:\Documents and Settings\DreamMaker>
```

Command Line interface

Offers **direct access** to system functionality
(e.g. CTRL+C)

Powerful and flexible
(batch, parameters)

Suitable for
repetitive tasks



Command Line Interface

C:\Windows\system32\cmd.exe

C:\>dir Temp

Il volume nell'unità C non ha etichetta.

Numero di serie del volume: 2E8A-3D64

Directory di C:\Temp

03/04/2013	16.30	<DIR>	.
03/04/2013	16.30	<DIR>	..
03/04/2013	16.28		0 due.txt
03/04/2013	16.28		0 uno.txt
		2 File	0 byte
		2 Directory	159.474.442.240 byte disponibili

C:\>ren "c:\Temp*.txt" *.doc

C:\>dir temp

Il volume nell'unità C non ha etichetta.

Numero di serie del volume: 2E8A-3D64

Directory di C:\temp

03/04/2013	16.31	<DIR>	.
03/04/2013	16.31	<DIR>	..
03/04/2013	16.28		0 due.doc
03/04/2013	16.28		0 uno.doc
		2 File	0 byte
		2 Directory	159.474.442.240 byte disponibili

Run



Type the name of a program, folder, document, Internet resource, and Windows will open it

Open:

cmd

OK

Cancel



Search Windows

Command Line Interface

Offers **direct access** to system functionality
(e.g. CTRL+C)

Powerful and flexible (batch, parameters)

Suitable for **repetitive tasks**

Using and **learning** commands is **difficult**

Command names/abbreviations should be meaningful in order to be **remembered!**

Better for expert users than novices



The WIMP Interface

Window Icon Menu Pointer
or
Window, Icon, Mouse and
Pull-down menu!

Default style for majority of
interactive computer systems,
especially PCs and desktop
machines

Widget: window object

Classic paradigms

- Command Line Interfaces (CLI)
- Graphical User Interfaces (WIMP)

Modern paradigms

- Post-WIMP (touch, gesture)
- Conversational interfaces
- Immersive interfaces (AR/VR)
- Ubiquitous / ambient computing



**SEE YOU IN
THE NEXT
UNIT**

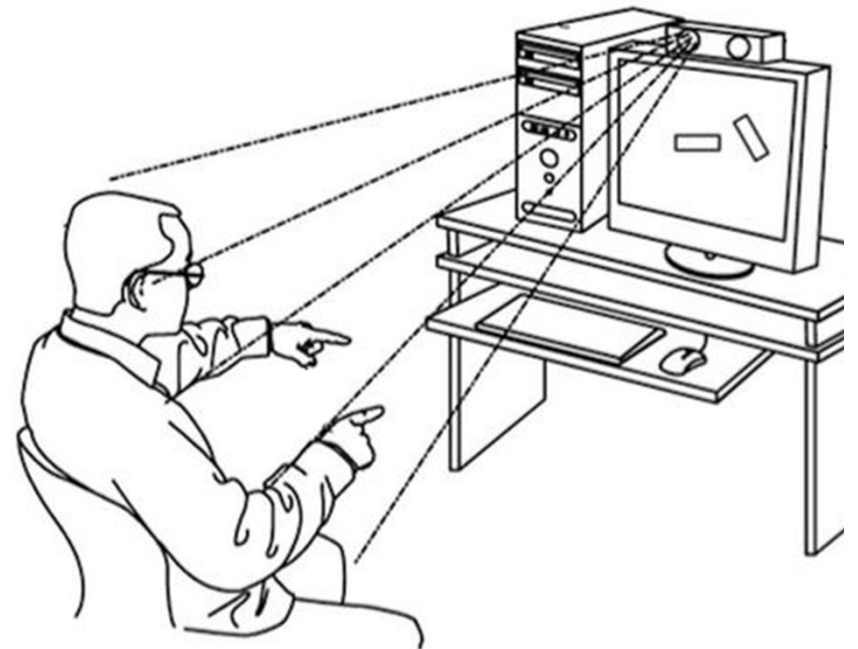
Modern Paradigms: Gesture Interfaces

Gesture recognition enables humans to **communicate with the machine and interact naturally through physical movements.**

Users interact by: hand gestures, touch gestures, full-body movements

Often used with: touchscreens
cameras, motion sensors

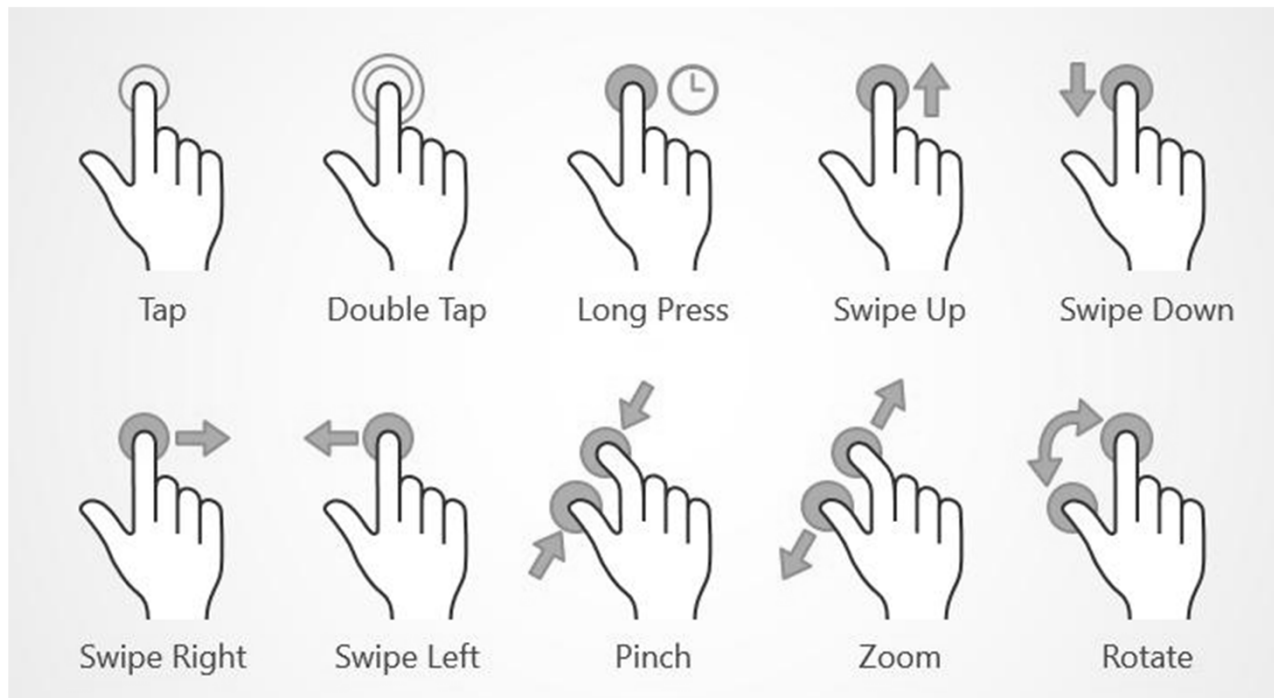
Gesture recognition can be conducted with techniques from **computer vision** and image processing.



Gesture Interfaces: Examples

smartphones (swipe, pinch, tap)
gaming systems (motion control)
smart TVs and AR/VR systems

Example device: Microsoft Kinect



The technology behind the Kinect



Gesture Interfaces: Pros and Cons

Advantages

natural interaction

intuitive for basic actions

no need for physical devices

Disadvantages

can be imprecise

difficult for complex tasks

may cause fatigue (“gorilla arm”)

Gesture Interfaces: standards?

Great potential but
demanding **new skills and standards**

Troubles have several causes:

- the lack of established guidelines for gestural control
- the misguided insistence by companies (e.g., Apple and Google) to ignore established conventions and establish ill-conceived new ones.



Conversational Interfaces

Conversational interfaces:
interaction through **natural language**

Users communicate via:
text
voice

System responses
simulate human dialogue.
So interaction becomes more
flexible and expressive, compared
to traditional interfaces.



Conversational Interfaces: Examples

Chat-based systems (where users type messages)

Example: ChatGPT

Voice assistants

(where users speak)

Examples: Siri, Alexa

These systems are now integrated into many devices, like smartphones, smart speakers, and cars.



Conversational Interfaces: Pros and Cons

Advantages

natural and intuitive

flexible input

no need to learn commands

Disadvantages

ambiguity in language

errors in understanding

lack of transparency.

syntax

e.g. the boy hit the dog with
the stick

meaning of words:

e.g. pitch

Conversational vs GUI: When to Use What

Conversational interfaces

open-ended tasks

information seeking

complex queries

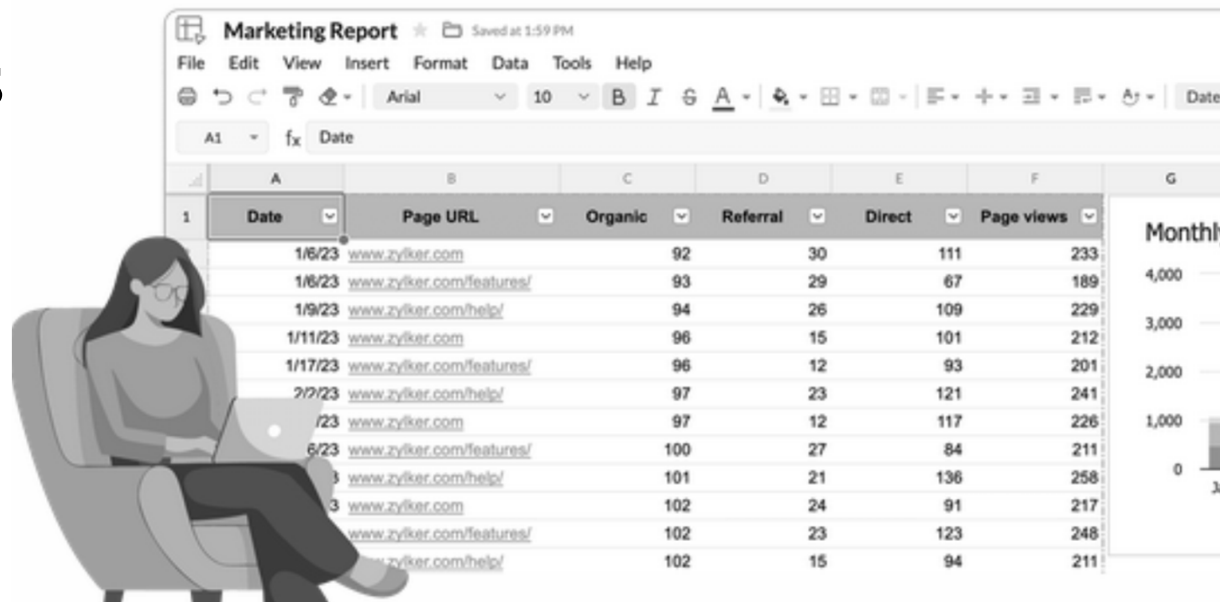
Graphical interfaces (GUI)

structured tasks

repetitive actions

precise control

Key idea: no single paradigm fits all tasks!



Immersive Interfaces (AR/VR)

Immersive interfaces: interaction **within a virtual or augmented environment**

Two main types:

- **Virtual Reality (VR)** → fully virtual world
- **Augmented Reality (AR)** → digital elements over real world

Goal:
increase sense of presence



Immersive Interfaces: Examples

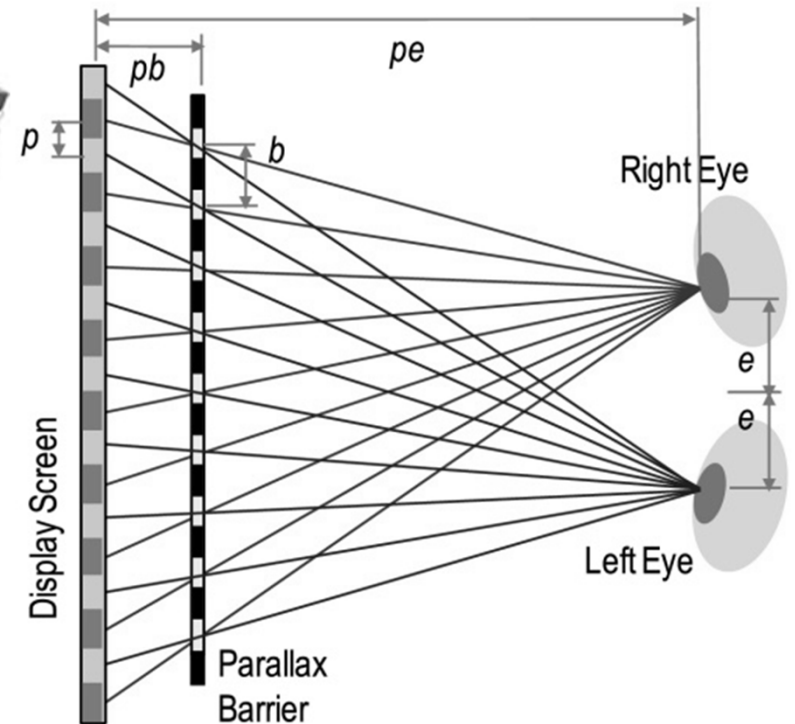
Examples:

- **VR headsets for gaming and training**
(e.g. Meta Quest)
- **AR applications on smartphones**
(e.g. Microsoft HoloLens)
- **industrial and medical simulations**



HW Example: VR with a monitor

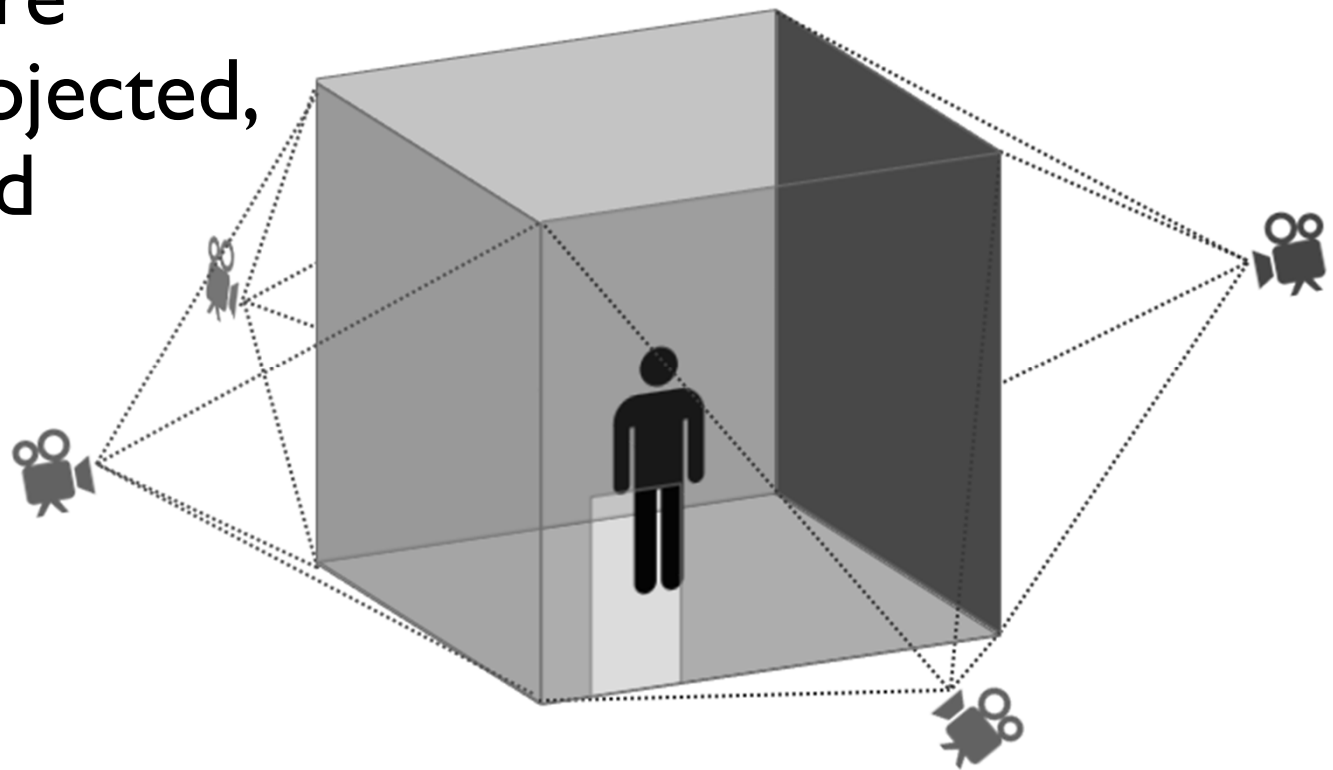
Commercially available **autostereoscopic displays** make 3D viewing without glasses feasible



HW Example: Surround-screen displays

Surround-screen displays, such as the CAVE™ (1992) are also popular. Instead of attaching the displays to the user, they place the displays in the world.

These displays are typically rear-projected, stereoscopic, and head-tracked.



Immersive Interfaces Pros and Cons

Advantages

- high level of engagement
- strong sense of presence
- effective for training and simulation

Disadvantages

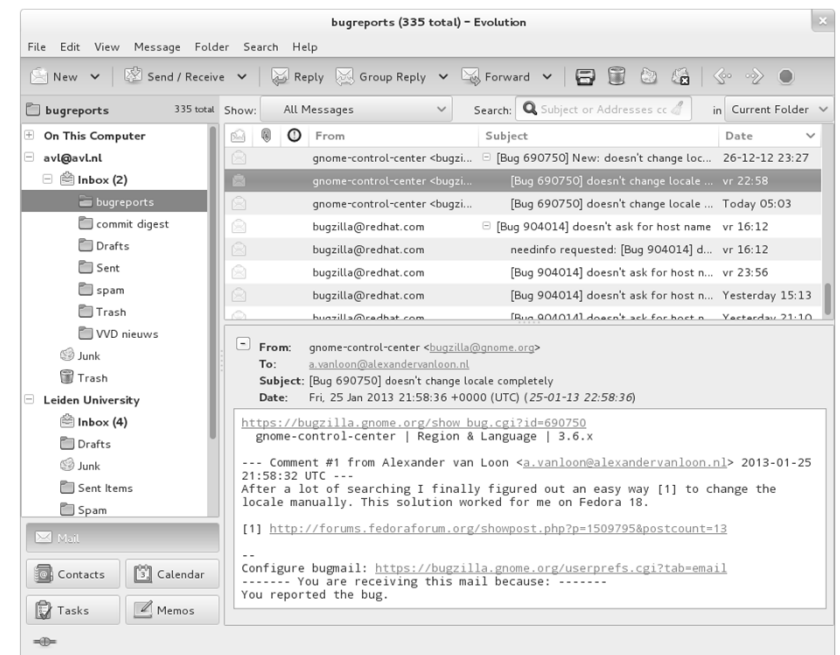
- expensive hardware
- physical discomfort (motion sickness)
- limited precision and usability

Why Immersive is not always better

More immersive does not mean better: it depends on the task.

For **simple tasks**, traditional interfaces are often **faster**.

Tasks that require precision, like writing or data entry, are usually better with traditional **interfaces**.



Immersive systems are most effective for **training**, **simulation**, and **experiential learning**.

Ubiquitous Computing

Ubiquitous computing: computation embedded in the environment

Characteristics:

- **invisible interaction**
- **always available**
- **context-aware systems**

Goal: interaction without explicit interfaces



How Ubiquitous Computing Is Achieved

The intention is to create a **computing infrastructure that permeates our physical environment** so much that **we do not notice the computer any longer.**

Two ways to make **computers disappear:**

- shrink and embed/distribute it in the **physical world**
- design interactions that **don't demand our intention**

Exploited **technologies** include wireless networking, voice recognition, camera and vision systems, pen-based computing and positioning systems..

Ubiquitous (or Pervasive) Computing

Small computers and sensors are embedded in various objects and places in our surroundings, and they communicate with each other and process information in a coordinated manner.



Designed by Ken Sakamura

Ubiquitous Computing: Context Awareness

Context awareness: the systems can **sense and understand** real world.

They **detect** location, user activity, environmental conditions.

They **adapt behaviour**, information, services!

So the system does not just react to commands, but it interprets the context and acts accordingly.

Ubiquitous Computing: Examples

Examples:

- **smart homes** (e.g. Google Home)
- **wearable devices** (e.g. Apple Watch)
- **smart cities** (traffic, energy, and services)



Ubiquitous Computing: Google Home

It works as a **central hub**, where users can configure, monitor, and automate different devices.

Instead of controlling each device separately, users interact with a **connected ecosystem**.

Ubiquitous Computing: Pros and Cons

Advantages

seamless interaction

(uninterrupted – invisible transitions)

personalized experience

reduced cognitive effort

Disadvantages

privacy concerns

loss of control and awareness

system opacity

system complexity

**IF THE INTERFACE DISAPPEARS,
WHO IS REALLY IN CONTROL?**



Interaction Paradigms: Summary

From explicit to invisible interaction:

CLI → commands and syntax

WIMP (GUI) → windows, icons, menus, pointer

Post-WIMP → touch and gestures

Conversational → natural language

Immersive (AR/VR) → interaction in 3D environments

Ubiquitous computing → interaction embedded in the environment